COMP8900
Jackie Ma
A00889988

Milestone 5

Per-File Hashing

Previous milestones hashed each watched directory as a single SHA-256 value. This detected when something changed but not which file was responsible. Implement per-file hashing. The agent now computes an individual SHA-256 hash for every file and stores results as a map of file path to hash value. The existing JSONB schema and change detection logic required no changes. The keys simply changed from directory paths to file paths.

A configurable file size threshold was added to skip large files such as media, which would otherwise cause the agent to hang. The default is 100 MB and is adjustable per-agent from the web UI. The frontend detail view was also redesigned as a timeline, where rows with changes expand to list the specific files added, modified or deleted.

Automated Testing

Unit and integration tests were written for the agent and backend, with a GitHub Actions workflow running them on every push. The Go tests cover skip path matching, hash correctness, large file filtering and change detection. The Python tests cover the change detection algorithm and all API endpoints against a real PostgreSQL database. The CI backend job provisions its own PostgreSQL container, keeping tests separate from the production Supabase instance.

The CI workflow runs both jobs in parallel. The backend job provides a PostgreSQL container using GitHub Actions' services block. This keeps the test database separate from the production Supabase instance.